

TRANSPLANTATION

Novel regulators of GVHD revealed through microbiome and metabolome patterns across distinct intestinal regions

Emma Lauder,^{1,*} Erik Anders Kiledal,^{2,*} Laure Maneix,³ Teal Furnholm,² Ana Santibanez,² Dongchang Zhao,³ Yaping Sun,³ Gregory J. Dick,² and Pavan Reddy³

¹Immunology Program and ²Department of Earth and Environmental Sciences, University of Michigan, Ann Arbor, MI; and ³Dan L Duncan Cancer Center, Baylor College of Medicine, Houston, TX

KEY POINTS:

- This study identified location-specific changes in the microbiome and associated gut metabolites in murine GVHD.
- This study identified phenyllactate as a novel regulator of GI GVHD.

Microbial dysbiosis and metabolite changes in the gastrointestinal (GI) tract have been linked to pathogenesis and severity of many diseases, including graft-versus-host disease (GVHD), the major complication of allogeneic hematopoietic stem cell transplantation. However, published studies have only considered the microbiome and metabolome of excreted stool and do not provide insight into the variability of the microbial community and metabolite composition throughout the GI tract or the unique temporal dynamics associated with different gut locations. Because such geographical variations are known to influence disease processes, we used a multi-omics approach to characterize the microbiome and metabolite profiles of gut contents from different intestinal regions in well-characterized mouse models of GVHD. Our analysis validated analyses from excreted stool, but importantly, uncovered new biological insights from the microbial and metabolite changes between syngeneic and allogeneic hosts that varied by GI location and time after transplantation. Our integrated analysis confirmed the involvement of known metabolic pathways, including short-chain fatty acid synthesis and bile acid metabolism, and identified additional functional genes, pathways, and metabolites, such as amino acids, fatty acids, and sphingolipids, linked to GI GVHD. Finally, we validated a biological relevance for one such newly identified microbial metabolite, phenyl lactate, that heretofore had not been linked to GI GVHD. Thus, our analysis of the geographic variability in the intestinal microbiome and metabolome offers new insights into GI GVHD pathogenesis and potential for novel therapeutics.

Introduction

The gut microbiome is composed of a diverse array of microbes that contribute to health and disease.¹ Disrupting the balance of the gut microbiota, dysbiosis is characterized by an increased abundance of facultative anaerobes and aerotolerant microbes.² Gut dysbiosis has been linked to many diseases, such as inflammatory bowel disease, obesity, diabetes, cancer pathogenesis, and responses to therapy, including graft-versus-host disease (GVHD), the major complication of allogeneic hematopoietic cell transplant (allo-HCT).²⁻⁴

In addition to direct interactions between the gut microbiome and host, gut metabolites are also critical regulators of intestinal health and diseases. Gut metabolites can be derived from host extrinsic sources, such as diet and microbiome, or from host intrinsic sources that are further processed or used by the gastrointestinal (GI) microbiome. Dysbiosis of the gut

microbiome leads to significant changes in the composition of gut metabolites that have profound impact on host health and disease severity.^{5,6}

Allo-HCT is a potentially curative therapy for patients with several hematological and inherited diseases. Its efficacy is limited by its most significant complication, the allogeneic T-cell-mediated GVHD. GI manifestation of GVHD causes significant morbidity and mortality.⁷

Several clinical and experimental studies have demonstrated that GVHD severity can be affected by changes in the stool microbiome and microbial metabolites, with dysbiosis correlated to worsened outcomes in patients with allo-HCT. Specifically, clinical studies have demonstrated that reduced gut microbial diversity, stemming from a loss of obligate anaerobes and increase in aerotolerant organisms, correlates with GVHD severity outcomes.⁸ Emerging data have also correlated GI

microbial metabolites, especially short-chain fatty acids and bile acids, with GVHD outcomes.^{5,9–11} Several microbiota-linked interventions (fecal-microbiota transplant, prebiotics, and probiotics) are currently in clinical trials to mitigate GVHD.^{12,13} However, these experimental and clinical studies have relied on analyses of excreted stool samples and did not assess microbiome and metabolome variability, dynamics, and specificity at different gut locations and their relevance to disease progression and severity.

Microbial communities found throughout the intestines differ from stool-associated communities based on geographical location.^{7,14,15} This intestinal geographical variability of the microbiome has been shown to correlate with the type and site of intestinal inflammation and the development of colon cancer.^{16,17} However, whether the microbial metabolites also vary with the geographical variation of the GI microbiome and whether this pattern changes in immune-mediated GI diseases remain unexplored. Specifically, it is unknown if GVHD-associated dysbiosis is observed across the intestinal anatomic landscape, whether this leads to geographic variation of microbial metabolites, and finally whether this variation is associated with GI GVHD severity and mortality. Herein, we addressed these critical questions using well-established murine models of GVHD. We performed shotgun metagenomics and untargeted metabolomics on intestinal contents harvested from 4 distinct regions of the small and large intestines, determined the microbiome composition and functional potential, and integrated these results with metabolomic profiles. We identified metabolites and microbiome differences associated with GVHD at different sites in the GI tract. We further identify and demonstrate a critical role for a novel GVHD-related metabolite, phenyl lactate, in regulating GI GVHD severity.

Materials and methods

Mice

C57BL/6 (number 027, B6, H-2Kb) and BALB/c (number 028, H-2Kd) mice were purchased from Charles River Laboratories. Mice were maintained in specific pathogen-free conditions.

The Institutional Animal Care and Use Committee protocol number was AN-8909.

Hematopoietic cell transplantation

Transplantations were performed as previously described.¹⁸ Briefly, splenic T cells from BALB/c donors or C57BL/6 were enriched by using Pan-T microbeads (130-095-130, Miltenyi Biotec), and bone marrow was collected from the same donors. On day 0, 3×10^6 T cells and 5×10^6 bone marrow cells were injected intravenously into C57BL/6 recipient mice. Recipient mice received 10 Gy of irradiation on day –1. BALB/c recipient mice received 2×10^6 T cells and 5×10^6 bone marrow cells on day 0 and 8-Gy split-dose irradiation on day –1. The mice were randomly assigned to syngeneic or allogeneic groups in each experiment. For phenyllactic acid (PLA) treatment, mice received daily oral gavage of PLA (10 mg/kg) starting on day 0 transplant.

Sample collection

On day 7 or day 21 after cell transplantation, recipient mice were euthanized. The GI tract was removed, and the gut contents from the terminal ileum, cecum, transverse colon, and descending colon were collected. Gut contents were pooled for $n = 5$ mice that were housed in the same cage.

DNA extraction, sequencing, and quality control

DNA was extracted from gut contents using the Qiagen Mag-Attract PowerMicrobiome kit (catalog number 27500-4-EP) and quantified with the Quant-iT PicoGreen dsDNA Assay kit (catalog number P7589). A spike-in control consisting of an even mix of DNA from 3 organisms (ATCC MSA-1014) was added to constitute 0.05% of the DNA. Shotgun metagenomic sequencing libraries were prepared by the University of Michigan Advanced Genomics Core using the NEBNext Ultra II FS DNA Library Prep Kit (NEB, E7805) and sequenced using an Illumina NovaSeq instrument (300 cycles, producing 150-bp paired-end reads).

Metabolomics

Untargeted metabolite profiles were determined using reversed phase chromatography/quadrupole time of flight in positive and negative modes by the University of Michigan Metabolomics Core. Sample preparation included a 1-step liquid-liquid organic solvent extraction of sample material. Analytes measured are reported as peak area per sample. Metabolomics analyses were performed using Metaboanalyst 5.0.

MetaboAnalyst 5.0 analysis

Peak areas were uploaded as a .csv. Reads with relative standard deviations >30 were filtered out. Data were normalized by transforming, and auto-scaling was performed (mean centered and divided by the standard deviation of each variable). Fold change analysis was performed on nonnormalized data (before log transformation and auto-scaling) by calculating the ratio in between 2 group means (mean1/mean2). *t*-Tests were performed with the following parameters: group variance = equal, *P* value threshold = 0.05 (false discovery rate). Volcano plots use fold change and *t* test values and plot the values on the *x* and *y* axes, respectively. Metabolite set enrichment analysis was performed on differentially abundant metabolites, as determined by *t*-tests. Globaltest was used to test associations between metabolite sets and the outcome. Our metabolite set was compared against 80 metabolite sets based on human Kyoto Encyclopedia of Genes and Genomes (KEGG) metabolic pathways.

Bioinformatics and statistical analysis

More detailed methods can be found in the supplemental Methods (available on the *Blood* website).

Intestinal organoid culture

Intestinal crypt-derived organoid culture was adapted from previous reports.^{19,20} More detailed methods can be found in the supplemental Methods.

Results

Loss of microbial diversity is variable across the intestinal landscape in GI GVHD

Loss of diversity in the stool microbiome is well established in existing studies of GVHD.⁴ We therefore used a well-established model of GVHD and first assessed whether a similar loss of diversity was observed in distinct intestinal locations, and if it was similar in degree to findings from stool analyses. We harvested the gut contents from 4 intestinal locations (terminal ileum, cecum, transverse colon, and descending colon) from the recipient animals after either syngeneic (B6 → B6) or allogeneic (BALB/c → B6) bone marrow transplantation (BMT) (Figure 1A). GVHD is observed as early as day 7 and gradually worsens by day 21 in these models. Microbial community composition and metabolites were measured using shotgun metagenomics and untargeted mass spectrometry, respectively. The samples were stratified by experimental condition (syngeneic or allogeneic), gut location (terminal ileum, cecum, transverse colon, and descending colon), and days after transplant (7 or 21 days). The within-sample (α) microbial diversity was characterized by both Inverse Simpson Index and Shannon diversity. Stool from the descending colon is most similar to excreted stool as measured via quantitative polymerase chain reaction (qPCR) and bioinformatic comparisons between distinct data sets (supplemental Figure 1A-C). Therefore, we compare results from the descending colon stool samples with previous results from excreted stool studies. Consistent with previous reports,¹³ microbial diversity was lower in allogeneic mice compared with syngeneic mice in the stool harvested from descending colon based on Shannon diversity (Figure 1B-C). However, although there was a significant ($P < .05$) decrease in Shannon diversity for allogeneic samples based on a linear model, the same was not observed for the inverse Simpson diversity metric. By both metrics, microbial diversity was significantly lower in samples collected 21 days after transplant (also observed in certain t test comparisons) and—relative to transverse colon samples—lower in terminal ileum samples and higher in cecum samples. Furthermore, consistent with previous reports from stool analysis,¹³ allogeneic mice on both day 7 and day 21 after transplant had an increase in relative abundance of facultative anaerobes (eg, *Escherichia*, *Enterococcus*), a trend toward an increase in mucus-degrading bacteria (eg, *Akkermansia*), and a decrease of obligate anaerobes (eg, *Ruminococcus*) (Figure 1D-G). Changes in relative abundance of all bacterial taxa are shown in the heat map in Figure 1H. We also used a multifactor analysis model, MEFISTO, to analyze our metagenomics and metabolomics data. Factor 2 separated our data based on experimental condition (syngeneic vs allogeneic). Negative factor weights associated with syngeneic conditions were correlated with Lachnospiraceae and Turicibacteraceae, whereas allogeneic conditions (positive factor weights) were associated with Oscillospiraceae and *Escherichia* (Figure 1I). The association of *Escherichia* with allogeneic is consistent with the bloom of *Escherichia coli* we see in allogeneic samples (Figure 1D). In addition to confirming previous observed changes in microbiome composition, our analysis of distinct gut locations after transplant highlighted several other key findings: first, the largest difference in α -diversity was attributable to gut location, driven largely by the significantly lower diversity observed in the terminal ileum (Figure 1B-C). Gut location also

explained the most variance in community composition or β -diversity, followed by time after transplant (Figure 1J-L; supplemental Figure 1B). These data highlight the significant impact of gut geography on microbiome composition.

Microbial pathways associated with GVHD

Previous studies using 16S sequencing have focused on changes in microbial community composition, providing limited insight about changes in the abundance of functional pathways. Therefore, in addition to analyses of differences in community composition, we also evaluated functional differences represented in the microbiome across the varied intestinal locations following allo-HCT. Traits were predicted for each metagenome-assembled genome using functional gene annotations, revealing an increase in oxygen-tolerant organisms in allogeneic samples compared with syngeneic in all gut locations, correlating with a bloom of aerotolerant microbes (facultative anaerobic microbes) (Figure 2A). Furthermore, we observed a decreased abundance of butyrate biosynthesis genes on both day 7 and day 21 in samples harvested from the descending colon of the allogeneic animals when compared with syngeneic samples (Figure 2B), consistent with previous studies reporting a decrease in the short-chain fatty acid butyrate in GVHD.⁵ Additionally, we observed significant differences in the abundance of 3 other metabolic pathways not previously associated with allo-HCT. First, allogeneic samples had a decreased abundance of organisms with carbohydrate processing genes compared with syngeneic controls (Figure 2C). Second, organisms capable of reducing thiosulfate to sulfite via thiosulfate reductase were decreased in all experimental groups after transplant compared with the naïve controls. This decrease was maximal in the terminal ileum in allogeneic samples on day 21 (Figure 2D). Finally, analysis of differentially abundant genes in allogeneic samples revealed an increase in genes coding for siderophore-dependent iron import into the cell when compared with the syngeneic controls (Figure 2E). The increase of siderophore biosynthesis genes was largely attributable to the bloom of *E coli* on day 7 in allogeneic samples (Figure 2F).

We next sought to identify gene pathways that were enriched in the experimental condition and gut locations. To do so, we evaluated KEGG pathways with gene set enrichment analysis, with genes ranked by estimates from an ANOVA-Like Differential Expression Tool 2 (ALDEx2) generalized linear model of KEGG ortholog abundance (KEGG ortholog abundance ~ transplant type $\times$ time after transplant $\times$ gut section) (Figure 2G). Allogeneic transplant samples were enriched in pathways indicative of a dysbiotic microbiome, such as *E coli* biofilm formation, antimicrobial peptide biosynthesis, infectious diseases, lipopolysaccharide biosynthesis, and siderophore biosynthesis (Figure 2G; supplemental Figure 2A-B). Furthermore, allogeneic samples were depleted in pathways associated with typical GI functions, such as starch and sucrose metabolism, nucleotide metabolism, and pentose and glucuronate interconversions; quorum sensing and *Staphylococcus aureus* infection pathways were also depleted (Figure 2G; supplemental Figure 2). Compared with allogeneic cecum and allogeneic descending colon samples, allogeneic terminal ileum samples have ≈ 5 times more unique enriched pathways. These pathways include pathways associated with metabolic independence,²¹ such as amino acid metabolism, carbohydrate

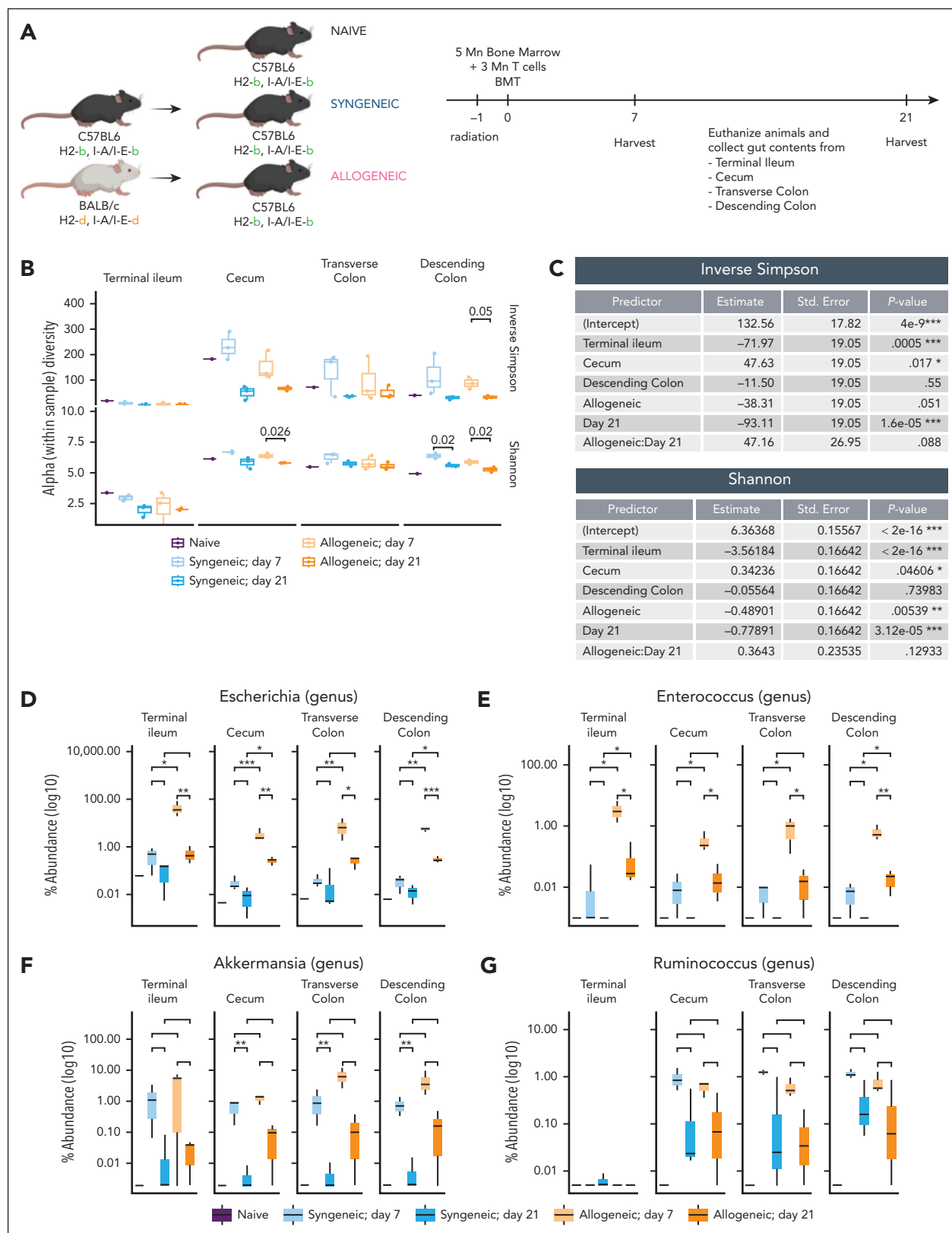


Figure 1. Microbial diversity varies with transplant type, gut section, and days after transplant. (A) Intestinal contents were collected from different gut locations from mice in 3 treatment groups; contents from those receiving syngeneic or allogeneic transplants were collected 7 and 21 days following transplant, and contents were also collected from naïve mice. (B-C) The microbial diversity in these samples was compared using Shannon and Inverse Simpson diversity indexes. (D-G) Significant negative effects on diversity were associated with allogeneic transplant samples, samples collected 21 days after transplant, and samples collected in the terminal ileum, whereas diversity was higher in cecum samples compared with transverse colon samples. Differential abundance of specific taxa was also evaluated. (D-E) *Escherichia* and *Enterococcus* were more abundant in samples from mice receiving allogeneic transplants regardless of gut section, and more abundant 7 days after transplant than 21 days after transplant.

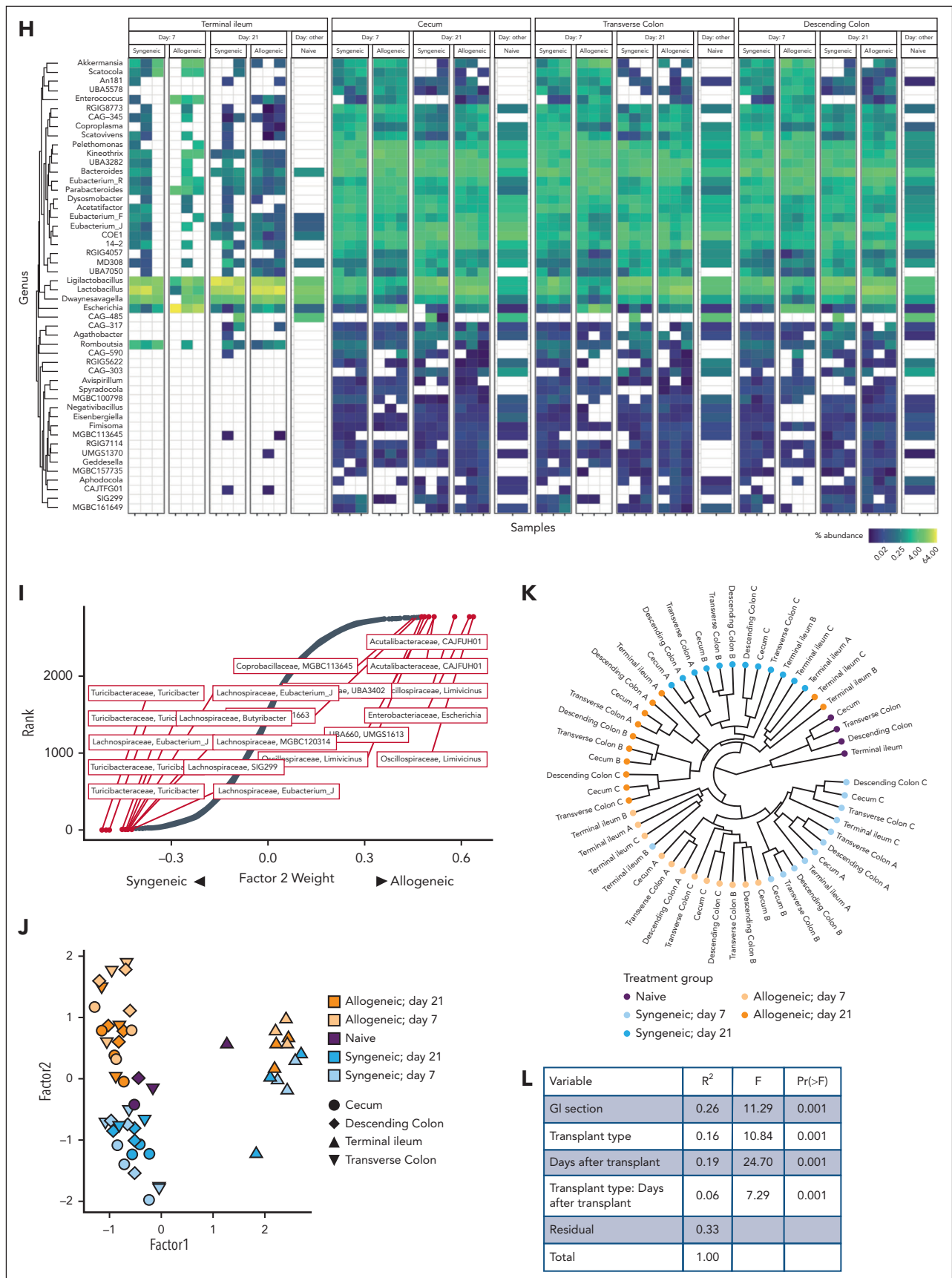


Figure 1 (continued) (F) *Akkermansia* were more abundant on day 7 in samples from both transplant types, with notably higher abundance in the colon samples of allogeneic transplant specimens from that day. (G) *Ruminococcus* were also more abundant in day 7 samples. (H) Heat map of the relative abundances of bacterial taxa. Factor analysis was conducted with MEFISTO, and 10 factors were calculated. Factor 1 separated samples based on gut location, with terminal ileum being an outlier. Factor 2 separated

metabolism, cell motility, membrane transport, and cofactor metabolism (Figure 2G; supplemental Figure 2). Additionally, allogeneic terminal ileum samples are enriched in the phosphotransferase transport system—important not only for sugar transport but also metabolic and virulence regulation²² (Figure 2G). The considerable increase in unique pathways associated with allogeneic terminal ileum samples highlights the importance of probing different gut locations and may provide additional insight into site-specific biology, such as enterohepatic biology.

Thus, these data collectively demonstrate that after allogeneic stem cell transplant, gut microbial diversity is reduced, and relative abundance of facultative anaerobes increases. Differences were observed in several microbial pathways with little previous association with GVHD, such as those involved in oxygen tolerance, carbohydrate metabolism, iron, and sulfur metabolism. We further observed geographic differences in pathway enrichment, indicative of increased pathogen colonization of the terminal ileum and cecum in allogeneic transplant recipients at days 7 and 21, respectively.

Paired unbiased metabolomics demonstrates variability of metabolites across intestinal landscape

We next evaluated whether the pathway differences observed in the metagenomic analysis correlated with the metabolite levels at the various intestinal sites after allo-HCT by performing untargeted metabolomic profiling of the paired samples with liquid chromatography–tandem mass spectrometry. A total of 139 metabolites were differentially abundant (t test with $P < .05$) between syngeneic and allogeneic samples on day +7 and day +21 after BMT. The transverse colon at day 21 after transplant had the most differentially abundant metabolites (Figure 3A), whereas the fewest differences were observed in the descending colon (which is most similar to excreted stool) on day 21 after allogeneic BMT transplant. This demonstrates that metabolites that are differentially abundant in other GI locations would not be recapitulated in excreted stool studies. Only 1 metabolite was differentially abundant in the allogeneic recipients in all 4 gut locations on day 7 after transplant: 2-acetylpyrrolidine. By contrast, on day 21 after transplant, 9 metabolites were differentially abundant at all gut locations, 6 of which were bile acids (Table 1). The low number of differentially abundant metabolites observed across all gut locations highlights the importance of probing different sites throughout the GI tract. Furthermore, in between day 7 and day 21 after transplant, the terminal ileum, cecum, transverse colon, and descending colon share 21, 5, 26, and 66 differentially abundant metabolites in a syngeneic vs allogeneic comparison, respectively (Table 2).

We next used MEFISTO to analyze the metabolites associated with syngeneic vs allogeneic conditions (Figure 3B) and

found new metabolites that are associated with syngeneic experimental conditions, including molecules present in cell membranes: (lysophosphatidylinositol [32:0] and monogalactosyldiacylglycerol [44:9]). However, this factor analysis did not subset the results based on different gut locations. Therefore, we next performed pathway enrichment analyses on metabolome changes from each gut location. The analyses demonstrated that the pathways involving bile acid biosynthesis and tryptophan metabolism were dysregulated at all intestinal locations (Figure 3C). Consistent with previous reports, all bile acids were decreased in the intestinal contents of allogeneic animals (supplemental Figure 3B).¹¹ Tryptophan metabolites indolelactic acid and 5-hydroxyindoleacetic acid were most different in between day 7 and day 21 (supplemental Figure 3C-D). Notably, we observed several differences that heretofore have not been reported: specifically, the most dysregulated pathways included those involved in energy metabolism (malate-aspartate shuttle, mitochondrial β -oxidation of long-chain fatty acids, glutamate metabolism, and glucose-alanine cycle), amino acid metabolism (valine, leucine, and isoleucine degradation, and alanine metabolism), and pathways involved in fatty acid metabolism (α linolenic and linoleic metabolism, fatty acid biosynthesis) (Figure 3C; supplemental Figure 4A-H). These observations suggest that pathways essential to cell homeostasis are dysregulated after allotransplant.

Acetylcarnitines and fatty acids were significantly increased on day 7 in the allogeneic samples in the terminal ileum and transverse colon, respectively (Figure 3D-E). Interestingly, in contrast to the increase in allogeneic samples for most fatty acids identified, only azelaic acid—an antimicrobial metabolite—decreased on day +7 in allogeneic samples. Additional key metabolic pathways, such as those involving sphingolipids and lysophospholipids, showed differentially abundant metabolites after transplant. Sphinganine, a key sphingolipid, was significantly increased on day +7 only in the transverse colon (Figure 3F), whereas a decrease in the levels of lysophospholipids (LPCs) was observed in allogeneic animals on both on days +7 and +21 when compared with syngeneic controls in the terminal ileum (Figure 3G). Finally, the metabolite phenyl lactate and its critical microbial regulator the *Lactobacillus* species showed maximal difference in the allogeneic animals (Figure 3H-I).

Identification and validation of a microbial metabolite with previously unidentified biological function after allo-HCT

Given that the multi-omic and multi-location analysis of microbiome and metabolites identified novel changes in pathways and metabolites, we next determined whether any of them had a reproducible impact on the biology and outcome of GVHD. To this end, we focused on the metabolite with the largest differences observed in the analysis described above, PLA,

Figure 1 (continued) samples based on experimental condition (syngeneic vs allogeneic). (I) Bacterial taxa associated with syngeneic samples is associated with negative factor 2 numbers, whereas taxa associated with allogeneic samples is correlated with positive factor 2 numbers. (J) Factor 1 and factor 2 were plotted for all samples. (L) Pairwise Bray-Curtis dissimilarities were calculated between samples, and PERMANOVA analysis showed that 26% of the community variation between samples was associated with gut section, 18% with days after transplant, and 16% with transplant type. (K) Hierarchical clustering of samples based on the Bray-Curtis dissimilarities revealed similar patterns, with samples generally clustering by replicate, then gut section, treatment, and day. Results from t test comparisons are shown ($*P \leq .05$; $**P \leq .01$; $***P \leq .001$); ALDEx2 was also used to evaluate differential abundance (results not shown in figure).

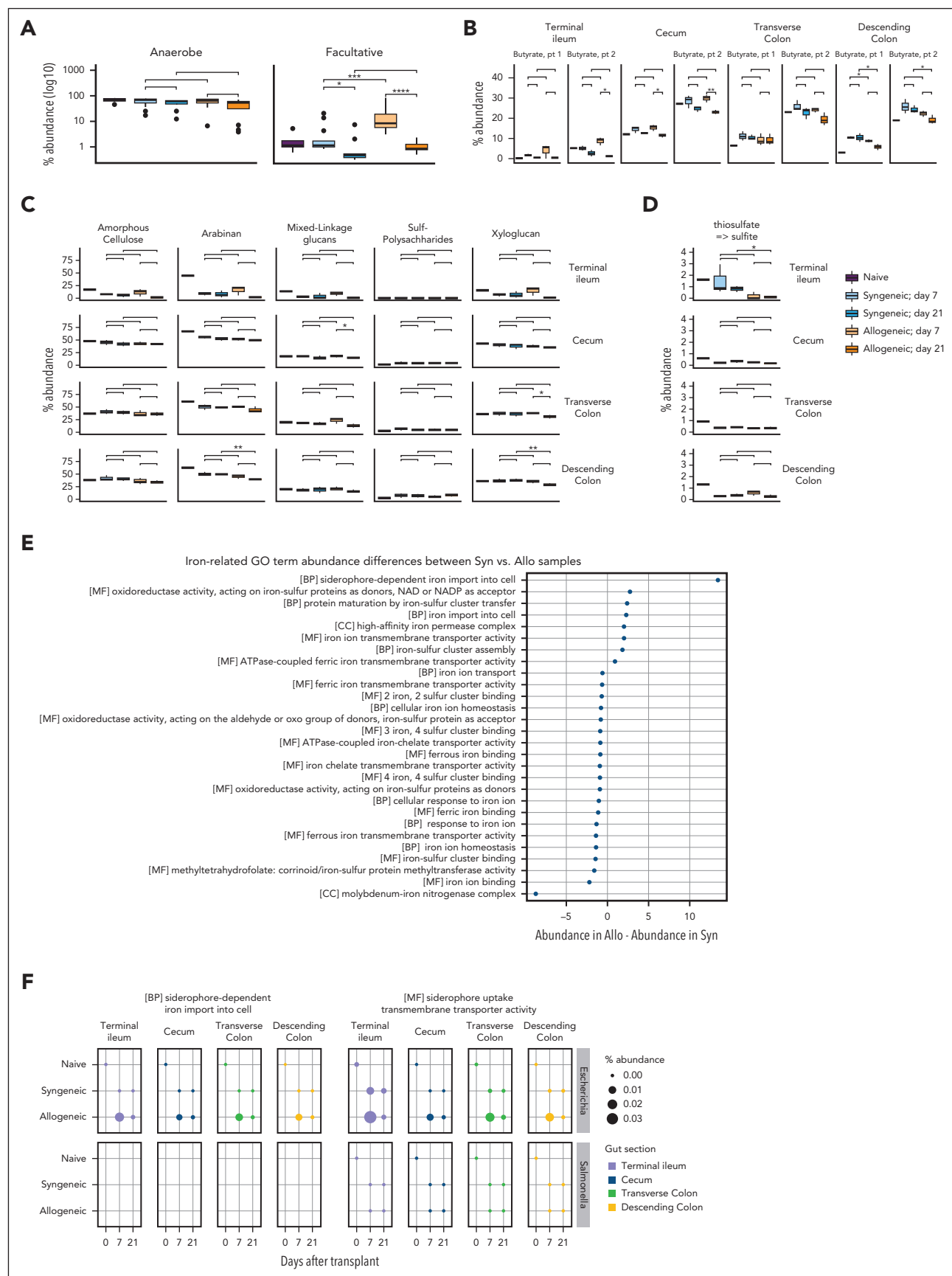


Figure 2. Microbial traits and functions differed in abundance across experimental groups. The oxygen tolerance of microbial community members was predicted and scaled by their abundance in each sample, revealing an increase in the abundance of oxygen-tolerant microbes in day 7 samples, particularly in samples from mice that

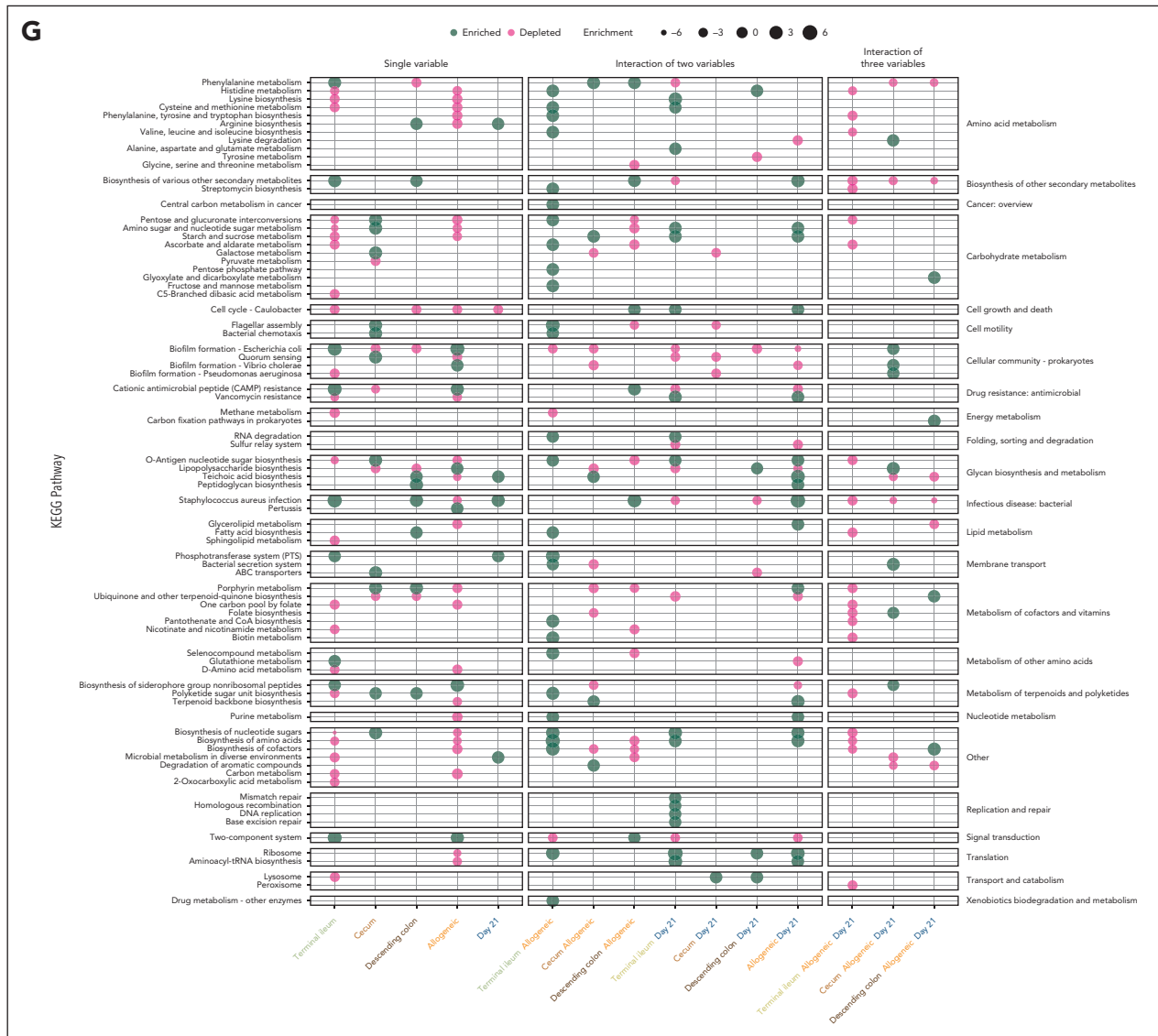


Figure 2 (continued) received allogeneic transplants. (A) The abundance of anaerobes also decreased across all transplant samples relative to naïve controls, and was lowest in allogeneic samples 21 days after transplant in samples from all gut locations. The abundance of key functional pathways inferred with the Distilled and Refined Annotation of Metabolism, such as butyrate production (phosphate butyryltransferase [patient (pt) 1] and terminal reactions [pt 2]) (B), carbohydrate degradation (C), and sulfur metabolism (D), was also evaluated; only pathways with at least 1 observation above a minimum abundance cutoff (1%) are shown. (B) Predicted butyrate producers were more abundant on day 7 in both transplant groups, and most abundant in cecum samples. (C) Carbohydrate degraders were most abundant in the lower GI samples with less consistent effects across treatment groups. Increased abundance of siderophore production and other iron-related gene ontology (GO) terms was observed in allogeneic samples, particularly those 7 days after transplant (E), with much of this difference attributed to the increased abundance of *E. coli* in those samples (F). (G) Gene set enrichment analysis was used to evaluate KEGG pathway enrichment using coefficients for each KEGG ortholog determined with an ALDEx2 generalized linear model encompassing the complete experimental design and interactions between terms. In the model, the following were used as the reference levels: syngeneic (treatment type), day 7 (days after transplant), and transverse colon (gut section). Enriched pathways (y-axis) are indicated by circles scaled by the enrichment score, shown in green for enriched pathways and red for depleted. Single variable effects are shown in the left-most horizontal grouping with interaction effects to the right. (A-D) Results from t test comparisons are shown (* $P \leq .05$; ** $P \leq .01$; *** $P \leq .001$; **** $P \leq .0001$); ALDEx2 was also used to evaluate differential abundance (results not shown in figure).

which was observed at higher concentrations in the descending colon of allogeneic animals (Figure 4A). Consistent with the increase in PLA, *Lactobacillus* species, which are a major PLA producer, were also more abundant (Figure 4B). We next analyzed whether this increase in PLA has any biological and potentially clinical implications for GVHD severity. Because host intestinal stem cell-derived organoids correlate with severity of GI GVHD,^{23,24} we analyzed intestinal organoid culture growth to assess the functional impact of PLA on the host intestinal regeneration. Addition of PLA to intestinal organoid cultures

inhibited organoid budding compared with diluent and pH matched controls (Figure 4C). Consistent with its effects on host intestinal stem cells, oral gavage of PLA in mice after allogeneic stem cell transplantation worsened GVHD severity and increased mortality (Figure 4D; supplemental Figure 5A). The increased mortality seen with PLA administration was strain independent as the phenotype was observed in C57BL/6 and BALB/c recipients (Figure 4D; supplemental Figure 5A). To explore the mechanism by which PLA worsened GVHD mortality, we performed reverse transcription-qPCR of

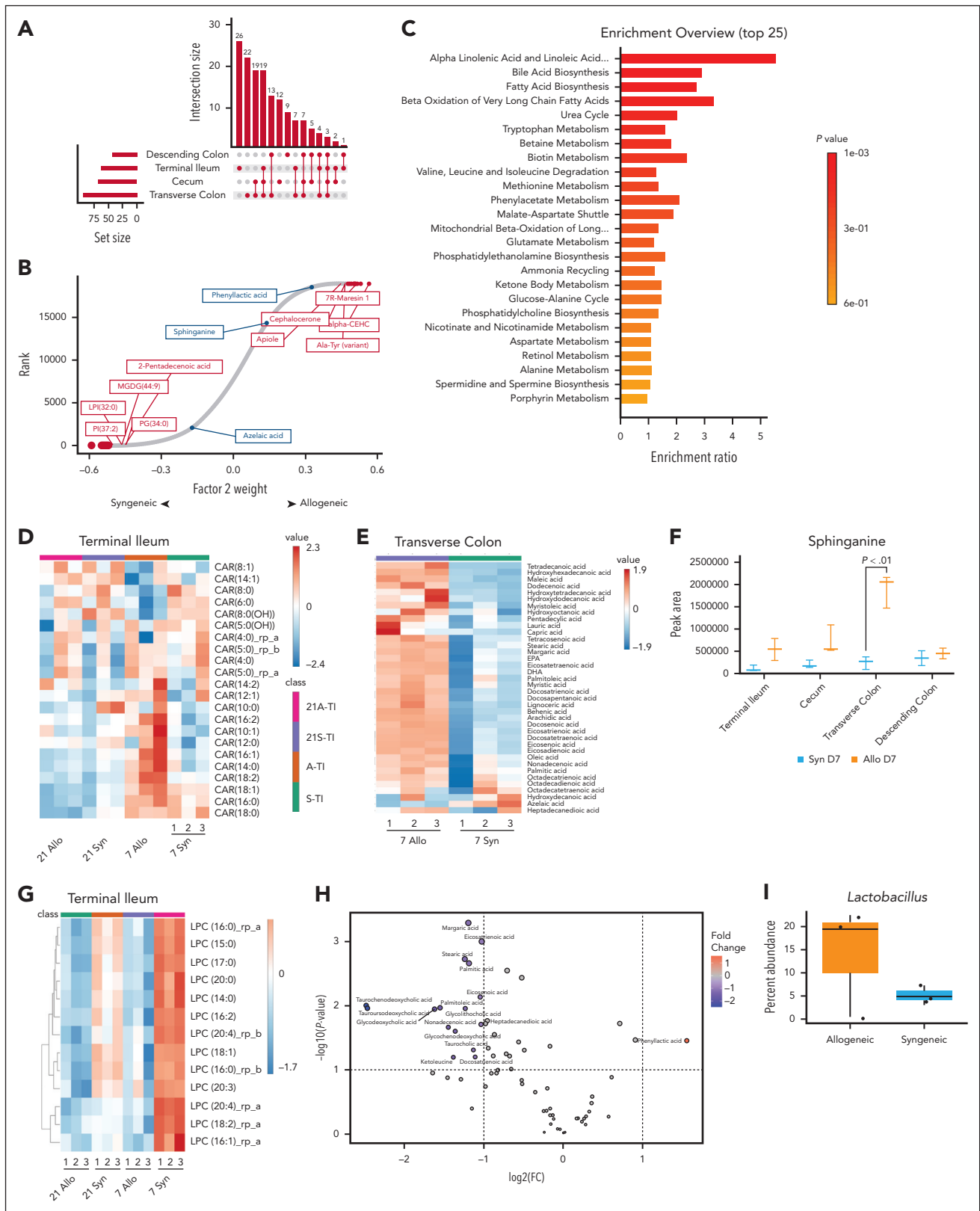


Figure 3. Metabolite differences associated with GVHD. The differentially abundant metabolites were identified by comparing syngeneic (Syn) and allogeneic (Allo) experimental groups within each gut location and time point after transplant (7 or 21 days after transplant) using the Metaboanalyst platform (Materials and methods). The overlap in significantly differentially abundant metabolites within each gut location and time point after transplant were compared to evaluate coordinated changes. The enrichment overview plots metabolites that were different (higher or lower in Syn vs Allo). (A) In this upset plot, points indicate group overlap, vertical bars indicate the number of shared metabolites, and horizontal bars indicate the number of metabolites differentially abundant between syngeneic and allogeneic samples in each gut location. Metabolite differences were most pronounced in the more proximal locations, with the most unique differentially abundant metabolites observed in the terminal ileum, and

Table 1. Differentially abundant metabolites shared between all 4 gut locations

Day 7 after transplant	Day 21 after transplant
2-Acetylpyrrolidine	2-Acetylpyrrolidine
	Glycochenodeoxycholic acid*
	Glycodeoxycholic acid*
	Ketoleucine
	Leu-Pro
	Lithocholic acid*
	Taurochenodeoxycholic acid*
	Taurohyodeoxycholic acid*
	Ursodeoxycholic acid*

*These are bile acids.

inflammatory cytokines (interferon gamma, interleukin 1 β [IL-1 β], and IL-17a) on the spleen and intestine after transplant. Additionally, we performed qPCR of select microbes from stool in the descending colon. Administration of PLA increased IL-1 β and IL-17a, but not interferon gamma, levels in the intestine and spleen. qPCR of microbes in the descending colon showed no differences between groups (Figure 4E-F). These data demonstrate identification, validation, and a functional role for a microbial metabolite, PLA, not previously implicated in regulation of GI GVHD.

Discussion

Many previous studies have analyzed microbiome and metabolite changes in GVHD from excreted stool or metabolite changes from patient serum.^{11,25,26} However, no study has analyzed the changes in microbiome and metabolome composition during GVHD at different gut locations. Herein, we performed such analysis and identified several novel changes in pathways and metabolites associated with GVHD and further demonstrated a critical biological and clinical role for a metabolite that has previously not been recognized to play a role in intestinal health.

Changes in the gut microbiome after allogeneic transplant, notably dysbiosis and altered gut metabolites, have been linked to worsened GVHD outcomes and GVHD gut pathology.^{2,6,8} Our microbiome profiling from different gut locations shows that the characteristics of a dysbiotic microbiome are present throughout the GI tract (terminal ileum to descending colon).

However, the changes in metabolites are largely specific to particular gut locations. This may be because of the strong role of geography in host physiology of absorption and excretion of metabolites, in addition to changes in microbiome composition. The lack of shared differentially abundant metabolites thus highlights the strong impact of gut location and disease severity (ie, time point after transplant) on gut metabolites. Our study also identified several other novel features in GI luminal metabolome following intestinal inflammation. All gut locations and time points after transplant shared only 1 differentially abundant metabolite in common: 2-acetylpyrrolidine. It was less abundant in allogeneic samples across all gut locations and all time points. This metabolite is found in human cereal products and could be found in the mouse feed; however, 2-acetylpyrrolidine has not been quantified in any mouse feed.²⁷ The decrease in 2-acetylpyrrolidine seen in allogeneic samples at all time points and at all gut locations is perhaps likely due to the allogeneic mice consuming less food after BMT.

Our study demonstrated that several other previously unconsidered metabolic pathways, such as acetylcarnitine metabolism, sphingolipid metabolism, fatty acid metabolism, and LPCs, are also disrupted by intestinal inflammation. Previous studies have shown that amino acids and lipids in the serum of patients are dysregulated pretransplant and reflect a risk of developing GVHD after allo-HCT.²⁸ The same metabolites found to be dysregulated in the serum of pretransplant patients are identified in the gut lumen of mice after transplant. Furthermore, low levels of LPCs have been previously linked to worsened prognosis in patients with GVHD, and LPCs have been shown to improve jejunum morphology, indicating identification of critical pathways that have been unrecognized by previous studies.^{29,30} Another intriguing and significant observation is that, although on day +7 after transplant there was a strong increase in multiple subtypes of fatty acids, only 1 fatty acid, azelaic acid, had the opposite trend and showed a decrease in allogeneic samples. Interestingly, azelaic acid has antimicrobial properties and is widely used in dermatology.^{31,32} Furthermore, recent studies have shown that oral gavage of azelaic acid ameliorated the dextran sulfate sodium-induced colitis phenotype.³³ However, the cause of reduced azelaic acid, its impact on the microbiome and the host, and its direct effect on GVHD severity or other intestinal inflammatory conditions will need to be addressed in future studies.

The analysis of functional pathways of the microbial community showed an increase in siderophore-dependent iron import in allogeneic samples on day +7 linked to a bloom of *E coli*. Siderophores are bacterial iron chelators that allow bacteria to compete for iron, an essential micronutrient.³⁴ Bacterial siderophores have been shown to modulate the microbial

Figure 3 (continued) similar differences observed in the terminal ileum, cecum, and transverse colon. Factor analysis was conducted with MEFISTO, and 10 factors were calculated. Factor 2 separated samples based on experimental condition (syngeneic vs allogeneic). (B) Metabolites associated with syngeneic samples are associated with negative factor 2 numbers, whereas metabolites associated with allogeneic samples are correlated with positive factor 2 numbers. (C) Enrichment analysis was performed using the Metaboanalyst 5.0 platform for metabolites that were significantly different between syngeneic and allogeneic samples from each gut location to determine which pathways are most affected. Heat maps of the peak area of carnitines in the terminal ileum (D), fatty acids from the transverse colon on day 7 after transplant (E), and lysophosphatidylcholines from the terminal ileum (G). (F) Peak area of sphinganine on day 7 after transplant showing significant increase in allogeneic samples in the transverse colon. (H) Volcano plot of differentially abundant metabolites in the descending colon 21 days after transplant. (I) Percentage abundance of *Lactobacillus* species in syngeneic vs allogeneic samples. LPI, lysophosphatidylinositol; MGDG, monogalactosyldiacylglycerol.

Table 2. Differentially abundant metabolites shared in between all day 7 and day 21 after transplant in each gut location

Terminal ileum	Cecum	Transverse colon	Descending colon
2-Acetylpyrrolidine	2-Acetylpyrrolidine	2-Acetylpyrrolidine	2-Acetylpyrrolidine
5'-Methylthioadenosine	2-Hydroxybutyric acid	2-Hydroxybutyric acid	Docosatetraenoic acid
CAR(8:0[OH])	CAR(14:1)	Arachidic acid	Eicosatrienoic acid
Glycochenodeoxycholic acid	Lithocholic acid	Behenic acid	Glycochenodeoxycholic acid
Kynurenic acid	Taurochenodeoxycholic acid	DHA	Octadecatetraenoic acid
Lithocholic acid		Docosapentaenoic acid	Palmitoleic acid
LPC(14:0)		Docosatetraenoic acid	
LPC(15:0)		Docosatrienoic acid	
LPC(16:0)_rp_a		Docosenoic acid	
LPC(16:0)_rp_b		Eicosadienoic acid	
LPC(17:0)		Eicosatetraenoic acid	
LPC(18:1)		Eicosatrienoic acid	
LPC(18:2)_rp_b		Eicosenoic acid	
LPC(20:0)		EPA	
LPC(20:3)		LPC(20:3)	
LPC(20:4)_rp_a		LPE(18:0)_rp_b	
LPC(20:4)_rp_b		Margaric acid	
LPE(18:2)_rp_b		Myristic acid	
SM(d40:2)		Myristoleic acid	
Sphinganine		Nonadecenoic acid	
Ursodeoxycholic acid		Palmitoleic acid	
		Pentadecylic acid	
		Ser-Leu	
		Stearic acid	
		Tetracosenoic acid	
		Ursodeoxycholic acid	

CAR, carnitine; DHA, docoheptaenoic acid; EPA, eicosapentaenoic acid; LPC, lysophospholipid; LPE, lysophosphatidylethanolamine; SM, sphingomyelin.

community and cellular iron homeostasis.³⁴ Previous studies have shown that administration of iron chelators reduced GVHD severity and, therefore, by limiting the amount of available iron, exogenous iron chelators could mitigate *E coli*'s competitive advantage and help restore eubiosis.³⁵ However, the impact of bacterial siderophores on dysbiosis and GVHD pathogenesis is unknown and requires further investigation. Furthermore, it remains unclear whether microbiome-derived siderophores regulate iron-dependent pathways, such as ferroptosis, in the host intestinal health and on immune responses.

Through metagenomic analyses, we observed an initial proliferation of *E coli* in samples collected 7 days after transplant and corresponding enrichment of pathways related to biofilm formation, siderophore biosynthesis, 2 component system, cationic antimicrobial peptide resistance, and secretion systems. These pathways were most enriched in day 7 allogeneic samples, and further enriched in the terminal ileum, highlighting important differences in the geography of microbial community responses to GVHD. Like the terminal ileum, flagellar assembly and chemotaxis pathways were enriched in the cecum: high levels of

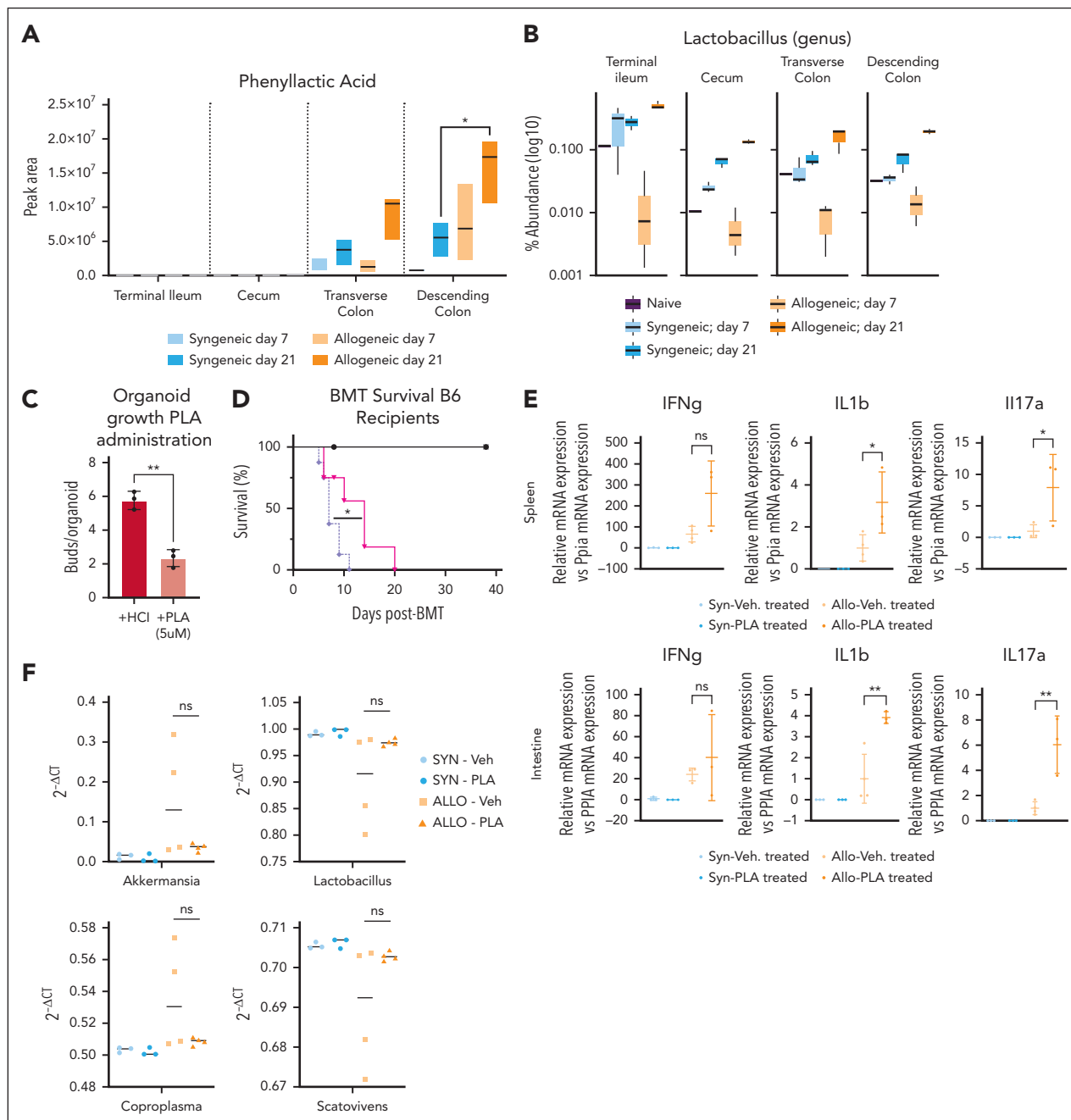


Figure 4. PLA and GVHD. (A) The peak area of PLA with 2-tailed t test ($*P < .05$), showing that PLA was most abundant in the lower GI locations, and more abundant in the allogeneic samples than syngeneic samples on day 21. (B) Bacteria in the *Lactobacillus* genus were also more abundant in the allogeneic samples on day 21. Intestinal organoids were cultured with either 5 μ M of D-phenyllactic acid or with culture media with HCl pH matched to 5 μ M of D-phenyllactic acid. (C) Budding and number of organoids were counted on day 5 after seeding, showing decreased budding in organoids treated with PLA (unpaired t test, $**P < .01$). (D) C57BL/6 recipients of syngeneic or allogeneic transplants received daily oral gavage of either vehicle (water) or PLA for the duration of the experiment. Allogeneic $n = 8$, syngeneic $n = 6$, 2 independent experiments. Log-rank (Mantel-Cox) test, $*P < .05$. (E) Reverse transcription-qPCR of the cytokines interferon gamma (IFN- γ), IL-1 β , and IL-17a was performed on spleen and intestine from mice 7 days after transplant receiving vehicle or PLA oral gavage (unpaired t test, $*P < .05$). (F) qPCR of *Akkermansia*, *Lactobacillus*, *Coproplasma*, and *Scatovivens* was performed on stool from descending colon from mice 7 days after transplant receiving vehicle or PLA oral gavage. ns, not significant.

flagellin, typically modulated by host adaptive immune responses, have been associated with inflammation and gut mucosal barrier breakdown.³⁶ Later in GVHD progression, we observed the cecum to be uniquely enriched in biofilm formation and siderophore biosynthesis pathways in allogeneic transplant recipients. Although we observed significant pathogen

colonization, such as the initial *E coli* bloom throughout the GI tract, it was more pronounced in proximal locations, like the terminal ileum and cecum, where we also observed enrichment of functions associated with pathogenesis. These results suggest that proximal gut locations may be most impacted by the effects of pathogenic bacteria in GVHD.

Integrating metabolomics data with microbiome data, we found that an increase in PLA correlated with a bloom of *Lactobacillus* on day +21 in allogeneic conditions compared with syngeneic conditions. Furthermore, addition of PLA to in vitro cultures resulted in decreased budding of organoids, and administration of PLA to mice after transplant decreased survival and increased inflammatory cytokine IL-1b and IL-17a in the spleen and intestine, suggesting that PLA has direct effects on the host and GVHD pathology. PLA is both a host and microbial-derived metabolite with antimicrobial properties, but the increased abundance of microbial producers, like *Lactobacillus*, suggests that production could at least be amplified by the changes in the structure of the microbiome after transplant.³⁷ PLA is used in the farming industry and is used as a “green antibiotic” in livestock feed.³⁸ Our work suggests that PLA levels go up in the context of intestinal inflammation, and increased levels of PLA impact host biology and GVHD survival. Because PLA also has antimicrobial properties, it likely also impacts the microbial community structure. PLA administration did not significantly change the abundance of select microbes in the descending colon, but the impact on microbes that we have not specifically analyzed cannot be ruled out. The specific and direct effect of PLA on the host without any confounding effects by the microbiome will have to be studied by using germ-free hosts in the future. Furthermore, the specific source of PLA in the context of GI inflammation (host or microbiome derived) and the mechanism by which it affects the host cellular subsets (immune cells, intestinal epithelial cells) will require further investigation.

Our study has explored the changes in microbiome and gut metabolites along the GI tract during GVHD, and several correlations have been made between microbiome and gut metabolite changes. However, disentangling the mechanism behind these changes is complex as the abundance of gut metabolites can be affected by changes in either production or consumption of said metabolite by both the microbiome and the host. Therefore, our study identifies several new avenues for future exploration. Furthermore, there are both quantitative and qualitative changes in the microbes and metabolites between the syngeneic and allogeneic animals. Therefore, the qualitative effects of the metabolites on the allogeneic animals may be distinct from the syngeneic animals. The distinction and the degree of impact caused by the qualitative effects vs the quantitative effects of the metabolites between the groups will need to be carefully analyzed in the future studies.

Our study, even if the most comprehensive analysis to date, has limitations. For example, parameters outside of disease progression can impact the changes observed, such as reduced food intake by mice with GVHD. Additionally, mice undergoing BMT (both syngeneic and allogeneic) consume acid water, akin to decontamination by oral antibiotics of hematopoietic stem cell transplant patients, which can also impact the microbiome. However, both reduced food intake and antibiotics are clinically relevant to patients undergoing allogeneic hematopoietic stem cell transplant. Another limitation of the study is that functional metagenomics does not translate to active transcripts or to protein expression. Therefore, metagenomics, even when complemented by metabolite data as it was here, does not fully

recapitulate the active microenvironment of the host cells and microbiome.

Our study identified several novel pathways and metabolites associated with GVHD, including the identification of PLA and validation of its effect on GVHD biology. Furthermore, this work has opened many future avenues for exploration and sets the stage for future studies to disentangle the complex but significant interactions on host health, including GVHD outcomes.

Acknowledgments

The authors acknowledge use of the Microbiome Core, Advanced Genomics Core, and germ-free facility of the University of Michigan's Biomedical Research Core Facilities for the preparation of samples.

This work was supported by the National Heart, Lung, and Blood Institute, National Institutes of Health grant P01HL149633 (P.R. and G.J.D.) and by the Cancer Prevention and Research Institute of Texas RR220033 (P.R.).

Authorship

Contribution: E.L. designed, conceived, and performed the experiments, provided intellectual input, analyzed and interpreted the data, and wrote the manuscript; E.A.K. provided intellectual input, analyzed and interpreted the data, and wrote the manuscript; T.F. analyzed the data; L.M. and A.S. performed and helped with experiments; D.Z. and Y.S. helped with experiments; G.J.D. provided intellectual input and edited the manuscript; and P.R. conceived and designed the project and experiments, provided intellectual input, and wrote and edited the manuscript.

Conflict-of-interest disclosure: The authors declare no competing financial interests.

ORCID profiles: E.L., [0000-0001-6619-4336](https://orcid.org/0000-0001-6619-4336); E.A.K., [0000-0003-4565-2187](https://orcid.org/0000-0003-4565-2187); L.M., [0000-0003-0044-5395](https://orcid.org/0000-0003-0044-5395); G.J.D., [0000-0001-7666-6288](https://orcid.org/0000-0001-7666-6288); P.R., [0000-0001-7264-8429](https://orcid.org/0000-0001-7264-8429).

Correspondence: Pavan Reddy, Dan L Duncan Comprehensive Cancer Center, Baylor College of Medicine, 1919 Old Spanish Trail, Houston TX 77054; email: pavan.reddy@bcm.edu.

Footnotes

Submitted 1 July 2024; accepted 27 December 2024; prepublished online on *Blood* First Edition 26 February 2025. <https://doi.org/10.1182/blood.2024025924>.

*E.L. and E.A.K. contributed equally to this study.

Raw (unprocessed) metagenomic reads were deposited at the National Center for Biotechnology Information under BioProject PRJNA1120521. Analysis scripts are available on GitHub (github.com/Geo-omics/GVHD_mouse_metagenomes).

The online version of this article contains a data supplement.

There is a [Blood Commentary](#) on this article in this issue.

The publication costs of this article were defrayed in part by page charge payment. Therefore, and solely to indicate this fact, this article is hereby marked “advertisement” in accordance with 18 USC section 1734.

REFERENCES

1. Byndloss MX, Pernitzsch SR, Bäuml AJ. Healthy hosts rule within: ecological forces shaping the gut microbiota. *Mucosal Immunol*. 2018;11(5):1299-1305.
2. Tilg H, Zmora N, Adolph TE, Elinav E. The intestinal microbiota fuelling metabolic inflammation. *Nat Rev Immunol*. 2020;20(1):40-54.
3. Köhler N, Zeiser R. Intestinal microbiota influence immune tolerance post allogeneic hematopoietic cell transplantation and intestinal GVHD. *Front Immunol*. 2018;9:3179.
4. Rafei H, Jenq RR. Microbiome-intestine cross talk during acute graft-versus-host disease. *Blood*. 2020;136(4):401-409.
5. Mathewson ND, Jenq R, Mathew AV, et al. Gut microbiome-derived metabolites modulate intestinal epithelial cell damage and mitigate graft-versus-host disease. *Nat Immunol*. 2016;17(5):505-513.
6. Yue X, Zhou H, Wang S, Chen X, Xiao H. Gut microbiota, microbiota-derived metabolites, and graft-versus-host disease. *Cancer Med*. 2024;13(3):e6799.
7. Ferrara JLM, Smith CM, Sheets J, Reddy P, Serody JS. Altered homeostatic regulation of innate and adaptive immunity in lower gastrointestinal tract GVHD pathogenesis. *J Clin Invest*. 2017;127(7):2441-2451.
8. Jenq RR, Ubeda C, Taur Y, et al. Regulation of intestinal inflammation by microbiota following allogeneic bone marrow transplantation. *J Exp Med*. 2012;209(5):903-911.
9. Haring E, Uhl FM, Andrieux G, et al. Bile acids regulate intestinal antigen presentation and reduce graft-versus-host disease without impairing the graft-versus-leukemia effect. *Haematologica*. 2021;106(8):2131-2146.
10. Lindner S, Miltiadous O, Ramos RJF, et al. Altered microbial bile acid metabolism exacerbates T cell-driven inflammation during graft-versus-host disease. *Nat Microbiol*. 2024;9(3):614-630.
11. Michonneau D, Latis E, Curis E, et al. Metabolomics analysis of human acute graft-versus-host disease reveals changes in host and microbiota-derived metabolites. *Nat Commun*. 2019;10(1):5695.
12. Riwe MM, Golob JL, Magenau J, et al. Feasibility of a dietary intervention to modify gut microbial metabolism in patients with hematopoietic stem cell transplantation. *Nat Med*. 2023;29(11):2805-2813.
13. Fredricks DN. The gut microbiota and graft-versus-host disease. *J Clin Invest*. 2019;129(5):1808-1817.
14. Jin M, Shang F, Wu J, et al. Tumor-associated microbiota in proximal and distal colorectal cancer and their relationships with clinical outcomes. *Front Microbiol*. 2021;12:727937.
15. McCallum G, Tropini C. The gut microbiota and its biogeography. *Nat Rev Microbiol*. 2024;22(2):105-118.
16. Li J, Zhu Y, Yang L, Wang Z. Effect of gut microbiota in the colorectal cancer and potential target therapy. *Discov Oncol*. 2022;13(1):51.
17. Ruan W, Engevik MA, Spinler JK, Versalovic J. Healthy human gastrointestinal microbiome: composition and function after a decade of exploration. *Dig Dis Sci*. 2020;65(3):695-705.
18. Reddy P, Maeda Y, Liu C, Krijanovski OI, Korngold R, Ferrara JLM. A crucial role for antigen-presenting cells and alloantigen expression in graft-versus-leukemia responses. *Nat Med*. 2005;11(11):1244-1249.
19. Sato T, Vries RG, Snippert HJ, et al. Single Lgr5 stem cells build crypt-villus structures in vitro without a mesenchymal niche. *Nature*. 2009;459(7244):262-265.
20. Yilmaz OH, Katajisto P, Lamming DW, et al. mTORC1 in the Paneth cell niche couples intestinal stem-cell function to calorie intake. *Nature*. 2012;486(7404):490-495.
21. Watson AR, Füssel J, Veseli I, et al. Metabolic independence drives gut microbial colonization and resilience in health and disease. *Genome Biol*. 2023;24(1):78.
22. Deutscher J, Francke C, Postma PW. How phosphotransferase system-related protein phosphorylation regulates carbohydrate metabolism in bacteria. *Microbiol Mol Biol Rev*. 2006;70(4):939-1031.
23. Takashima S, Martin ML, Jansen SA, et al. T cell-derived interferon- γ programs stem cell death in immune-mediated intestinal damage. *Sci Immunol*. 2019;4(42):eaay8556.
24. Reddy P, Zhao D, Ravikumar V, et al. Inflammatory memory restrains intestinal stem cell regeneration. *Res Sq*. Preprint posted online 7 March 2023. <https://doi.org/10.21203/rs.3.rs-2566520/v1>
25. Thiele Orberg E, Meedt E, Hiergeist A, et al. Bacteria and bacteriophage consortia are associated with protective intestinal metabolites in patients receiving stem cell transplantation. *Nat Cancer*. 2024;5(1):187-208.
26. Vallet N, Salmona M, Malet-Villemagne J, et al. Circulating T cell profiles associate with enterotype signatures underlying hematological malignancy relapses. *Cell Host Microbe*. 2023;31(8):1386-1403.e6.
27. Human Metabolome Database. Showing metabocard for 2-acetylpyrrolidine (HMDB0037293). Accessed 2 May 2024. <https://hmdb.ca/metabolites/HMDB0037293>
28. Kumari R, Palaniyandi S, Hildebrandt GC. Metabolic reprogramming—a new era how to prevent and treat graft versus host disease after allogeneic hematopoietic stem cell transplantation has begun. *Front Pharmacol*. 2020;11:588449.
29. Liu Y, Huang A, Chen Q, et al. A distinct glycerophospholipid metabolism signature of acute graft versus host disease with predictive value. *JCI Insight*. 2019;5(16):e129494.
30. Li X, Abdel-Moneim AME, Mesalam NM, Yang B. Effects of lysophosphatidylcholine on jejuna morphology and its potential mechanism. *Front Vet Sci*. 2022;9:911496.
31. King S, Campbell J, Rowe R, Daly M, Moncrieff G, Maybury C. A systematic review to evaluate the efficacy of azelaic acid in the management of acne, rosacea, melasma and skin aging. *J Cosmet Dermatol*. 2023;22(10):2650-2662.
32. Layton AM, Dias da Rocha MA. Real-world case studies showing the effective use of azelaic acid in the treatment, and during the maintenance phase, of adult female acne patients. *Clin Cosmet Invest Dermatol*. 2023;16:515-527.
33. Liao Y, Wu X, Luo W, et al. Azelaic acid regulates the renin-angiotensin system and improves colitis based on network pharmacology and experimentation. *ACS Omega*. 2023;8(17):15217-15228.
34. Ellermann M, Arthur JC. Siderophore-mediated iron acquisition and modulation of host-bacterial interactions. *Free Radic Biol Med*. 2017;105:68-78.
35. Seike K, Kiledal A, Fujiwara H, et al. Ambient oxygen levels regulate intestinal dysbiosis and GVHD severity after allogeneic stem cell transplantation. *Immunity*. 2023;56(2):353-368.e6.
36. Cullender TC, Chassaing B, Janson A, et al. Innate and adaptive immunity interact to quench microbiome flagellar motility in the gut. *Cell Host Microbe*. 2013;14(5):571-581.
37. Phenyllactic acid: a potential antimicrobial compound in lactic acid bacteria. MedCrave online. Accessed 7 April 2023. <https://medcraveonline.com/JBMOA/phenyllactic-acid-a-potential-aantimicrobial-compound-in-lactic-acid-bacteria.html#ref7>
38. Rajanikar RV, Nataraj BH, Naithani H, Ali SA, Panjagari NR, Behare PV. Phenyllactic acid: a green compound for food biopreservation. *Food Control*. 2021;128:108184.

© 2025 American Society of Hematology. Published by Elsevier Inc. All rights are reserved, including those for text and data mining, AI training, and similar technologies.